Econ 109 Election Prediction Project

Presentation on October 27; write-up due on November 3 at 5pm

Each group is assigned a U.S. House or Senate race. If you do not yet have an assigned race, please reach out to the instructors as soon as possible.

Objective

The objective of the project is to use different sources of political and economic information to predict the outcome of your assigned race.

Your final prediction should include:

- the predicted winner;
- the predicted vote shares (or vote margin);
- an explanation of which pieces of evidence were most important for your prediction;
- a discussion of the main sources of uncertainty in your prediction.

The objective is not to build the most sophisticated prediction algorithm possible. Instead, the goal is to think carefully about what different datasets tell us about an election and how we can combine different pieces of evidence into a prediction.

Data collection and independent research

An important component of this project is learning how to find, evaluate, and work with data on your own.

The instructors will provide guidance and can help you troubleshoot problems, but we will not provide the datasets needed to complete the project. You should expect to search for appropriate data sources, determine how to download or extract the relevant information, and decide which variables are useful for your analysis. Each district and state presents its own data challenges. For instance, some districts have recently been redrawn, which can make collecting historical election results difficult. You may need to use results at the county or precinct level, or results from past elections under previous district boundaries. While these tasks may be challenging, we ask that you make every reasonable effort to resolve data issues on your own before requesting help from the teaching team.

The links provided at the end of this document are intended as starting points, not as an exhaustive list of sources. You are encouraged to search for other data that may be especially useful for your particular race.

Part of the exercise is deciding:

- which data are relevant to your question;
- what geographic and time units are appropriate;
- whether different sources can reasonably be compared or combined;
- what the limitations of each source are.

Do not worry if some of the data require cleaning or do not initially come in exactly the form you need. Dealing with these issues is part of empirical research. At the same time, you are not expected to undertake an excessively complicated data-construction exercise simply for its own sake.

Structure of the analysis

We suggest organizing your analysis in three steps.

Step 1: Construct a baseline prediction

Start from past election results to construct a simple baseline prediction for your race.

For example, you might consider:

- outcomes of previous elections in the district or state;
- recent presidential and congressional vote shares;
- whether previous elections indicate a persistent partisan advantage;
- whether the district or state has recently become more or less competitive;
- how turnout has changed across recent elections.

You will need to find and collect the relevant historical election data yourself.

Your baseline does not need to be complicated. Its purpose is to give you a starting point against which you can evaluate additional information.

Step 2: Bring in additional evidence

In addition to the past election results used to construct your baseline, your project should use the following three additional core data components:

1. **Demographics**

   Use Census or other demographic data to describe the electorate and investigate whether the district or state is changing over time.

   Possible variables include age, education, income, race and ethnicity, population growth, employment, occupation, urbanization, or other characteristics you believe are relevant.

2. **Campaign finance**

   Use campaign-finance data to study fundraising and contributions in the race.

   You might consider total receipts, cash on hand, expenditures, the timing of contributions, the number or type of contributors, outside spending, or other features that you think contain useful information.

3. **Polls**

   Collect available polling information for your race.

   Think carefully about:

   - when the poll was conducted;
   - the sample and sample size;
   - the reported margin of error;
   - whether different polls give similar or different signals;
   - how much weight you want to place on older versus more recent polls.

   You do not need to treat the latest poll as your prediction. One purpose of the project is to understand when other sources of information lead you to a prediction that differs from the polls.

In addition, each group should find at least one additional source of information that you think is particularly useful for understanding your race.

Possible examples include:

- candidate characteristics or previous political experience;
- incumbency;
- candidate policy positions;
- local economic conditions;
- turnout or voter-registration data;
- endorsements;
- campaign advertising or campaign activity;
- outside spending;
- characteristics of important local industries;
- local newspaper coverage;
- Google Trends or other measures of public attention;
- any other source that you can justify as informative.

This fifth component is deliberately open-ended. Part of the assignment is identifying a source of information that is particularly relevant to your race.

Step 3: Produce your final prediction

Use the evidence you have collected to move from your baseline prediction to your final prediction.

For each source of information, ask:

> *What does this information tell us that we did not already know from the baseline?*

You should be able to explain why each important piece of evidence moves your prediction—or why it does not.

You are not required to put every dataset into a single regression. Some sources may enter directly into a statistical model, while others may be used as descriptive or supporting evidence.

Your final analysis should make clear which pieces of information you found most informative and how you combined potentially conflicting signals.

Prediction methods

Students may use any prediction method they would like.

A simple linear regression is completely sufficient:

- `reg` in Stata;
- `lm` in R;
- `LinearRegression` in `scikit-learn` in Python.

You are welcome to experiment with methods such as Random Forest, XGBoost, or other prediction algorithms, but more complicated methods will not receive additional credit simply because they are more complicated.

A simple and well-understood model is preferable to a complicated model that you cannot explain.

Nor does every piece of evidence need to enter a formal prediction model. For example, you may use historical election results to construct a statistical baseline and then use polling, fundraising, or candidate information to evaluate whether that baseline should be adjusted.

The goal of the project is also not to obtain the "correct" prediction at all costs. We are interested in your reasoning, your use of evidence, and your understanding of uncertainty.

Presentation

Students will present their predictions in class on October 27.

Presentations will be approximately 3 minutes long. To keep the presentation focused, each group should prepare approximately three slides:

1. What is distinctive about the district or state and the candidates?
2. What are the most informative signals from your data? You do not need to present every variable or dataset.
3. What do you predict, how does your prediction compare with the polls, and what is the main source of uncertainty?

The presentation should include at least one clear visual representation of the evidence, such as a graph, map, or table.

There will be a panel of four faculty members and the two GSIs assessing the presentations and providing feedback. You should incorporate relevant feedback into the final write-up.

Write-up

The final write-up should explain your analysis and prediction in a way that could be understood by someone who has not seen your presentation.

It should contain:

1. a short description of the race and the district or state;
2. your baseline prediction based on the historical election data you collected;
3. a description of the four core data components and your additional data source, including where you obtained each dataset;
4. an explanation of what each important source of information adds to your prediction;
5. your final prediction;
6. a comparison between your prediction and available polls;
7. a discussion of the main sources of uncertainty and what could make your prediction wrong.

The write-up should be approximately 5–10 pages, including figures and tables.

For each dataset, briefly document its source and any important choices you made when constructing or cleaning it. We do not need a detailed account of every data-cleaning step, but someone reading the write-up should understand where the information came from and what it measures.

You should favor clear and informative visualizations over long collections of regression output. For example, if demographic change is important to your argument, show that change in a figure rather than simply reporting a list of coefficients.

Grading

Grading will be based on both the in-class presentation and the final write-up.

We will focus primarily on:

1. **Data collection and understanding:**

   Were you able to identify appropriate data sources? Do you understand what each dataset measures, its strengths and limitations, and why it might be informative about the election?

2. **Use and communication of evidence:**

   Are you able to identify the important signal in the data and communicate it clearly through graphs, maps, tables, or other visualizations?

3. **Reasoning behind the prediction:**

   Is it clear how you move from the different pieces of evidence to your final prediction? Can you explain why your prediction differs from your initial baseline or from available polls?

4. **Connection to concepts discussed in class:**

   Do you use concepts, models, and empirical reasoning from the course to interpret the evidence?

5. **Understanding of uncertainty:**

   Do you recognize the limitations of your analysis and identify what could make your prediction wrong?

The grade will depend on the quality of the analysis that produced the prediction.
Students should submit the code used for their analysis together with the written summary.

Use of AI

Students may use AI tools to help debug or check their code.

However, you should understand and be able to reproduce all of the analysis that you submit. You should be able to explain what every line of code used in your prediction model does and why you used it.

AI should not substitute for your own decisions about which data are informative, how the evidence should be interpreted, or why you arrive at your final prediction.

Potentially useful data resources

The sources below are intended to help you get started. You are not expected to limit yourself to these sources, and finding appropriate data on your own is an important part of the project.

Different races may have different useful sources. In particular, state and local election agencies, local newspapers, candidate websites, and state-specific databases may contain information that is not available in a single national dataset.

- **Past election results:**
  - MIT Election Data and Science Lab: `https://electionlab.mit.edu/data`
  - Historical federal election results: `https://www.usa.gov/election-results`
  - David Leip's Atlas of U.S. Presidential Elections: `https://uselectionatlas.org/RESULTS/`
- **Demographics and geographic data:**
  - Census data on IPUMS NHGIS: `https://www.nhgis.org`
  - Census TIGER/Line geographic files: `https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html`
  - County Business Patterns on local employment data by industry `https://www.census.gov/programs-surveys/cbp.html`
- **Campaign finance:**
  - FEC bulk data: `https://www.fec.gov/data/browse-data/?tab=bulk-data`
- **Polls:**
  - New York Times polling pages: `https://www.nytimes.com/interactive/polls/`

  - Local newspapers and television stations may also commission polls for individual House and Senate races

- **Candidates and issues:**
  - Vote Smart: `https://justfacts.votesmart.org/`
  - Ballotpedia: `https://ballotpedia.org/`
  - Candidate campaign websites
  - Incumbents' congressional websites and voting records on `https://voteview.com`

- **Other potentially useful sources:**
  - Google Trends: `https://trends.google.com/`
  - State election offices for voter-registration and turnout data
  - Bureau of Labor Statistics for local labor-market data: `https://www.bls.gov/`
  - Bureau of Economic Analysis for state and local economic data: `https://www.bea.gov/`
  - Prediction markets:
    * Kalshi election markets: `https://kalshi.com/category/elections`
    * Polymarket: `https://polymarket.com/politics`
    * PredictIt: `https://www.predictit.org`
  - Local newspapers and other local news sources
  - Campaign advertising, endorsements, and other information that you can justify as informative for your race

For campaign-finance data, note that the reporting period ending September 30 has an October 15 filing deadline. You may update your analysis when those reports become available.

Useful forecasting examples

It may be useful to look at how professional election forecasters combine different sources of information. These examples are intended as inspiration: you are not expected to reproduce their models.

- **Silver Bulletin from Nate Silver:** combines polling, leading indicators, historical information, and expert race ratings.
- **The Economist election model:** combines polls with fundamentals such as past election results, incumbency, fundraising, and candidate characteristics, and uses simulations to represent uncertainty.
- **Decision Desk HQ:** provides an example of an ensemble prediction model combining several statistical and machine-learning methods, including linear regression, regularized regression, Random Forest, and gradient boosting.
- **Cook Political Report / Inside Elections:** provide useful examples of expert-based race ratings that combine quantitative evidence with qualitative information about candidates and campaigns.

When looking at these forecasts, do not focus only on their final prediction. Try to understand:

- What data enter the forecast?
- What is the baseline prediction?
- How are polls incorporated?
- How are candidate- or race-specific factors incorporated?
- How does the model represent uncertainty?